

# Biomarkers of Kidney Failure and All-Cause Mortality in CKD

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## Key Points

- Established risk factor models demonstrate excellent discrimination for CKD outcomes but do not adequately reflect disease activity or mechanisms.
- We have developed models solely from novel biomarkers that reflect key disease pathways driving CKD progression with equivalent discrimination.
- Although biomarkers offer limited incremental gains in risk prediction, they may provide critical insights into disease mechanisms and treatment response.

## Abstract

**Background** CKD carries a variable risk for multiple adverse outcomes, highlighting the need for a personalized approach. This study evaluated several novel biomarkers linked to key disease mechanisms to predict the risk of kidney failure (first event of eGFR <15 ml/min per 1.73 m<sup>2</sup> or KRT), all-cause mortality, and a composite of both.

**Methods** We included 2884 adults with nondialysis CKD from 16 nephrology centers across the United Kingdom. Twenty-one biomarkers associated with kidney damage, fibrosis, inflammation, and cardiovascular disease were analyzed in urine, plasma, or serum. Cox proportional hazards models were used to assess biomarker associations and develop risk prediction models.

**Results** Participants had mean age 63 (15) years; 58% were male and 87% White. Median eGFR was 35 (25–47) ml/min per 1.73 m<sup>2</sup>, and the median urinary albumin-to-creatinine ratio was 197 (32–895) mg/g. During median 48 (33–55) months of follow-up, 680 kidney failure events and 414 all-cause mortality events occurred. For kidney failure, a model combining three biomarkers (soluble TNF receptor 1, soluble cluster of differentiation 40, and urinary collagen type 1  $\alpha$ 1 chain) showed good discrimination (C-index, 0.86; 95% confidence interval [CI], 0.83 to 0.89) but was outperformed by a model using established risk factors (age, sex, ethnicity, eGFR, and urinary albumin-to-creatinine ratio; C-index, 0.90; 95% CI, 0.88 to 0.92). For all-cause mortality, a model using three biomarkers (high-sensitivity cardiac troponin T, N-terminal pro-brain natriuretic peptide, and soluble urokinase plasminogen activator receptor) demonstrated equivalent discrimination (C-index, 0.80; 95% CI, 0.75 to 0.84) to an established risk factor model (C-index, 0.80; 95% CI, 0.76 to 0.84). For the composite outcome, the biomarker model discrimination (C-index, 0.78; 95% CI, 0.76 to 0.81) was numerically higher than for established risk factors (C-index, 0.77; 95% CI, 0.74 to 0.80), and the addition of biomarkers to the established risk factors led to a small but statistically significant improvement in discrimination (C-index, 0.80; 95% CI, 0.77 to 0.82; *P* value <0.01).

**Conclusions** Risk prediction models incorporating novel biomarkers showed comparable discrimination to established risk factors of kidney failure and all-cause mortality.

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## Introduction

CKD is an independent risk factor of kidney failure, cardiovascular events, and higher all-cause mortality.<sup>1</sup> Although the link between CKD and adverse outcomes is well established, the level of risk can vary significantly between individuals, highlighting the need to better understand the factors contributing to this heterogeneity to support more personalized approaches to care. While creatinine-based eGFR and urinary albumin-to-creatinine ratio (UACR) are validated as strong predictors of CKD outcomes<sup>2</sup> and are the basis of the risk-based Kidney Disease Improving Global Outcomes CKD classification system, as well as the kidney failure risk equation (KFRE),<sup>3</sup> they offer limited mechanistic insights into disease progression.

In recent decades, there has been a surge in research on serum, plasma, and urine biomarkers in CKD,<sup>4</sup> driven by a growing interest in identifying more specific and personalized predictors of adverse CKD outcomes. Large longitudinal CKD studies, such as the Canadian Study of Prediction of Death, Dialysis, and Interim Cardiovascular Events<sup>5</sup> and the Chronic Renal Insufficiency Cohort study,<sup>6,7</sup> have investigated the utility of novel biomarkers. Although several new biomarkers have been identified,<sup>8,9</sup> clinical uptake has remained low, partially because of failure to demonstrate significantly improved predictive accuracy over established models such as KFRE.<sup>10</sup> However, given the already strong performance of established risk factors in broad populations, the value of novel biomarkers may not be to improve risk prediction but rather to provide insights into CKD progression mechanisms to support personalized medicine, monitoring of responses to therapy, and novel drug development. Despite this promise, consensus on the clinically most useful biomarkers or combinations remains lacking.<sup>11</sup>

The National Unified Renal Translational Research Enterprise (NURTuRE)-CKD study sought to expand biomarker research by evaluating a large number of previously identified CKD biomarkers to determine the most effective combination for risk prediction and to gain deeper mechanistic insights.<sup>12</sup> Serum, plasma, and urine biomarkers were selected on the basis of their associations with pathophysiological processes, including kidney damage and/or fibrosis, inflammation, and cardiovascular disease, which are both drivers and consequences of CKD.<sup>13,14</sup> The aims of this study were to validate biomarkers previously investigated in large cohorts and to explore the potential of novel biomarkers linked to kidney failure and all-cause mortality in smaller studies.

## Methods

### Study Participants and Design

NURTuRE-CKD is an ongoing, prospective, UK-based multicenter cohort study of 2996 adults with an eGFR of 15–59 ml/min per 1.73 m<sup>2</sup>, or eGFR ≥60 ml/min per 1.73 m<sup>2</sup> and UACR >300 mg/g, recruited from 16

secondary care nephrology centers, along with 86 adult controls without CKD (23 with diabetes and 63 without diabetes). Exclusion criteria included solid organ transplant, undergoing dialysis, life expectancy of <1 year, AKI or major cardiovascular event within 3 months before enrollment, or current chemotherapy for cancer. Recruitment was completed between 2017 and 2019. Participants were registered with the UK Renal Registry for ongoing collection of local laboratory test results for up to 15 years. The UK Renal Registry also provides information on KRT initiation and deaths. Participants attended one follow-up visit, during which additional blood and urine samples were collected, and questionnaires were completed. Study design and cohort details have been reported previously.<sup>12</sup>

### Biomarker Selection

The biomarkers selected for assay were associated with three important broad mechanisms of CKD progression and/or premature death: kidney damage and/or fibrosis, inflammation, and cardiovascular disease. Eight urinary biomarkers ( $\alpha$ -glutathione S-transferase, osteopontin, thyroid transcription factor 3,  $\beta$ -2 microglobulin, urinary neutrophil gelatinase-associated lipocalin [NGAL], urinary cystatin C, uromodulin, and EGF) were excluded after initial selection because of either inadequate assay precision (intra- and/or interassay variation coefficient >20%) or unsuitability of uncentrifuged urine for their measurement. Uncentrifuged urine was selected for analysis because it more closely reflects clinical practice in routine urine testing. Four serum and plasma biomarkers—IL-17, IL-1, IL-6, and TNF $\alpha$ —were also excluded because of inadequate assay precision. The final panel consisted of 21 biomarkers: 13 measured in blood (EDTA or lithium heparin plasma and serum) and eight measured in uncentrifuged urine (U). Of these, 12 were primarily related to kidney damage/fibrosis, five to inflammation, and four to cardiovascular disease (Supplemental Table 1).

### Sample Collection and Laboratory Measurement

Biomarkers were measured centrally at Geneva University Hospitals, Switzerland. Samples were collected according to a standard operating procedure that required plasma and serum samples to be frozen within 2 hours of collection. Serum creatinine, C-reactive protein (CRP), growth differentiation factor 15 (GDF-15), N-terminal pro-brain natriuretic peptide (NT-proBNP), high-sensitivity cardiac troponin T (hs-cTnT), and UACR were analyzed on routine Roche cobas 8000 modular analyzer series. Except for GDF-15, all these analyses are International Organization for Standardization 15189 certified. Plasma (lithium heparin) soluble cluster of differentiation 40 (sCD40), soluble cluster of differentiation 40 ligand (sCD40L), fibroblast growth factor 23, galectin 3, kidney injury molecule-1 (KIM-1), monocyte chemoattractant protein-1 (MCP-1), NGAL, and soluble TNF receptor 1 (sTNFR1) were measured according to the manufacturer's

instructions on a Meso Scale Discovery platform, a solid-phase multiplex electrochemiluminescent immunoassay. Urine biomarkers measured on the Meso Scale Discovery plate comprised calbindin, KIM-1, clusterin, osteonectin, and vascular endothelial growth factor (VEGF). Plasma (EDTA) soluble urokinase plasminogen activator receptor (suPAR) was measured by a Virogates immunoassay according to manufacturer's instructions. Urinary tissue inhibitor of metalloproteinases-1 (UTIMP-1), urinary matrix metalloproteinase-9 (UMMP-9), and urinary collagen type 1  $\alpha$  1 (UCOL1A1) were measured on commercially available ELISAs (Biotechne) according to manufacturer's instructions. Intra- and interassay precision data are presented in Supplemental Table 2. Baseline UACR and serum creatinine were measured centrally, and GFR was estimated using the 2009 CKD Epidemiology Collaboration equation without the ethnicity variable as recommended by the National Institute for Health and Care Excellence in 2021.<sup>15</sup> Routine biochemistry analyses and full blood counts were performed in local hospital laboratories.

#### Dataset and Preprocessing

The NURTuRE-CKD cohort consisted of 2996 participants with CKD and 86 controls. Fifteen participants with CKD were excluded because of a lack of baseline samples, and four participants with CKD withdrew their consent. A further 93 participants had an eGFR  $<15$  ml/min per  $1.73$  m<sup>2</sup> at centralized assessment and were excluded before analysis, resulting in a final CKD cohort of 2884 participants. Of note, four of the 21 biomarkers, specifically clusterin, tissue inhibitor of metalloproteinases-1, COL1A1, and MMP-9, had values out of the working range of the assays in over 10% of samples, either above the upper limit of quantification (ULOQ) or below the lower limit of quantification (LLOQ). To address this, the assay's lower limit was assigned to values below the LLOQ and the upper limit was assigned to values above the ULOQ. Supplemental Table 1 summarizes all missing values for each biomarker and %  $>$ ULOQ and  $<$ LLOQ.

#### Definitions of Clinical Variables and Outcomes

Comorbidities were assessed by questionnaire and confirmed using medical records. Ethnicity was self-reported. Primary kidney diagnosis was obtained from medical records and categorized using European Renal Association-EDTA codes.<sup>16</sup>

The clinical end points were as follows:

- **Kidney failure:** the first instance of eGFR falling below  $15$  ml/min per  $1.73$  m<sup>2</sup>, sustained for at least 28 days, or the initiation of KRT (dialysis or transplant), whichever occurred first.
- **All-cause mortality:** death from any cause occurring before the development of kidney failure.
- **Composite of kidney failure or all-cause mortality:** either of the two outcomes.

Participants who had not reached an end point were censored on December 31, 2022.

#### Statistical Methods

##### Data Transformation

Urinary biomarkers were normalized for urinary creatinine concentration and are presented as a ratio to urinary creatinine, used in all statistical models. Biomarker data underwent logarithmic transformation before analysis, and results are presented per SD higher to allow for more intuitive comparison across biomarkers.

##### Statistical Association Tests

Differences between groups were assessed using Mann-Whitney test for continuous variables and log-rank test for categorical variables. A  $P$  value  $< 0.05$  was considered statistically significant. Spearman rank correlation coefficients were calculated to evaluate associations between biomarkers and results displayed using heatmaps.

##### Statistical Modeling: Cox Proportional Hazards Analysis

Cox proportional hazard models were developed to investigate the relationship between the biomarkers and time to end points. Hazard ratios (HRs) and 95% confidence intervals (CIs) were calculated per SD higher biomarker level. Hierarchical models were constructed: unadjusted associations; model 1—sex, age, and White ethnicity; model 2—additionally adjusted for UACR and eGFR. For the end point of all-cause mortality and the composite end point, an additional model was added, model 3, additionally adjusting for the index of multiple deprivation score, smoking, alcohol, diabetes, hypertension, prior vascular disease (a composite of coronary revascularization procedures, myocardial infarction, stroke, peripheral vascular disease, and amputation), statins, and angiotensin-converting enzyme inhibitor or angiotensin receptor blocker use. The biomarker associations with each outcome were then subanalyzed by the eGFR stratum:  $>45$ ,  $30-45$ , and  $<30$  ml/min per  $1.73$  m<sup>2</sup>.

To assess the potential for nonlinear associations between biomarkers and outcomes, we performed a sensitivity analysis using natural cubic spline functions within the Cox proportional hazards models. Missing values were handled using variable type-specific imputation. For continuous variables, the median of observed values was used, and for categorical variables, the mode.

##### Stepwise Biomarker Combination for Risk Prediction

To support the identification of the biomarkers most relevant to CKD progression and mortality, biomarkers were ranked on the basis of their discrimination for the three outcomes within the Cox proportional hazards model using the C-index statistic. To establish the best performing combination, biomarkers were iteratively added to the model according to C-index ranking until saturation was achieved—i.e., until adding another biomarker did not improve the model. The most parsimonious model was chosen, optimizing the balance between simplicity and performance by prioritizing fewer biomarkers with minimal C-index loss. The dataset was randomly split into training and testing sets using an 80:20 ratio, separately for the three end points to ensure independence of the testing set. Bootstrap resampling (1000 iterations) was used to calculate 95% CIs for the C-index on both training and

testing datasets. To understand whether adding the parsimonious biomarker model to an established risk factor model resulted in improved discrimination, the compareC version 1.3.2 R package, which accounts for censored observations when comparing concordance indices, was used. To obtain diagnostic accuracy measures—including sensitivity, specificity, negative predictive value, and positive predictive value—we computed area under the receiver-operating characteristic curves using the pROC package (version 1.18.4) in R. The DeLong method was used to compare models.

Analyses were performed using python (version 3.8.10), R (version 4.3.1), and Jupyter Notebook (version 5.2.0).

## Results

### Baseline Characteristics

Baseline data for the CKD cohort, subdivided according to end point, are presented in Table 1. Mean (SD) age was 63 (15) years; 58% were male and 87% White. Median (25th–75th percentile) eGFR was 35 (25–47) ml/min per 1.73 m<sup>2</sup>, and median UACR was 197 (32–895) mg/g. The mean age for the control group was 56 years, with 37% male and 27% had diabetes.

The median biomarker concentrations across GFR categories of CKD and in control groups are provided in Table 2. Five biomarkers (urinary KIM-1, suPAR, urinary clusterin [UClusterin], sCD40, and GDF-15) were significantly higher in the controls with diabetes than those without diabetes. All biomarkers, except for sCD40L, were higher in participants with CKD than in controls; only sCD40L and urinary matrix metalloproteinase 9 did not show higher values across GFR categories. Biomarker concentrations stratified by end point are provided in Supplemental Table 3. Supplemental Figure 1 is a heatmap representing biomarker correlations and highlighting clustering between biomarker mechanisms.

### Biomarker Associations with Kidney Failure

During a median (25th–75th percentile) follow-up of 48 (33–55) months, 680 participants progressed to kidney failure. A total of 629 participants reached an eGFR of <15 ml/min per 1.73 m<sup>2</sup>, while an additional 51 participants initiated KRT (42 started dialysis and nine received kidney transplants) before reaching an eGFR of <15 ml/min per 1.73 m<sup>2</sup>. Associations with baseline biomarker concentrations adjusted for clinical covariates (age, sex, White ethnicity, baseline eGFR, and UACR) are shown in Figure 1A. Among the kidney damage/fibrosis biomarkers, plasma KIM-1, suPAR, UCOL1A1, and UClusterin showed positive associations with kidney failure, while urinary calbindin (UCalbindin), UMMP-9, and VEGF (UVEGF) were inversely associated. The inflammatory biomarkers sCD40, MCP-1, and sTNFR1 were associated with a higher risk of kidney failure. Two of the cardiovascular disease biomarkers, hs-cTnT and GDF-15, were also associated with a higher risk of kidney failure. For full modeling results, see Supplemental Table 4.

When analyzed by eGFR strata, most biomarkers significantly associated with kidney failure remained significant in two of the three individual eGFR strata. Notably,

UClusterin and UCalbindin were significant in only those with an eGFR <45 ml/min per 1.73 m<sup>2</sup> and UVEGF, hs-cTnT, and inflammatory biomarkers sCD40, and sTNFR1 were only significant in those with an eGFR <30 ml/min per 1.73 m<sup>2</sup>. See Supplemental Tables 5–7.

### Biomarker Associations with All-Cause Mortality (before Kidney Failure)

There were 414 all-cause mortality events before kidney failure in the CKD cohort. Nine kidney damage/fibrosis biomarkers (KIM-1 [serum and urine], NGAL, suPAR, UClusterin, urinary osteoactivin [UOsteoactivin], UCalbindin, UVEGF, and UTIMP-1), four inflammatory biomarkers (sCD40, sTNFR1, CRP, and MCP-1), and all four cardiovascular disease biomarkers (hs-cTnT, NT-proBNP, GDF-15, and fibroblast growth factor 23) were associated with higher risk of all-cause mortality in the fully adjusted model (see Figure 1B and Supplemental Table 8). When analyzed by eGFR strata, most of these biomarkers remained significantly associated with all-cause mortality. UCalbindin, UVEGF, and UTIMP-1 were only significant in the 30–45 ml/min per 1.73 m<sup>2</sup> subgroup, while NGAL was only significant in the most advanced group. See Supplemental Tables 9–11.

### Biomarker Associations with the Composite End Point of Kidney Failure or All-Cause Mortality

Eight of the kidney damage and fibrosis biomarkers—plasma and urine KIM-1, NGAL, suPAR, UClusterin, UOsteoactivin, UCOL1A1, and UTIMP-1—along with four inflammatory biomarkers—sCD40, sTNFR1, CRP, and MCP-1—and all four cardiovascular biomarkers—were associated with the composite end point of kidney failure or mortality in the fully adjusted model (see Figure 1C, and Supplemental Table 12). When analyzed by eGFR strata, most of these biomarkers remained significantly associated with the composite outcome. Notably, NGAL, UVEGF, and MCP-1 were significant only in the most advanced stage, while urinary KIM-1, UTIMP-1, UOsteoactivin, and UCalbindin were significant in the two earlier stages but not in the most advanced stage. See Supplemental Tables 13–15.

Our sensitivity analysis has shown that on the basis of Akaike information criterion and Bayesian information criterion comparisons, model fit was not meaningfully improved with spline-based models compared with linear models for outcomes (see Supplemental Tables 16 and 17). Therefore, linear-based modeling is presented here.

### Combination Biomarkers for the Discrimination of Kidney Failure and All-Cause Mortality

The unadjusted C-index, ranked for each biomarker for the end point of kidney failure, all-cause mortality, and the composite of the two, is displayed in Figure 2, A–C, respectively. Full results from the training and testing sets are presented in Supplemental Tables 18–20 and comparison of training and testing set characteristics in Supplemental Tables 21 and 22. Iterative addition of

**Table 1. Summary of baseline demographic and clinical variables in 2884 participants with CKD**

| Baseline Demographics                | Whole Cohort (n=2884) | Kidney Failure (n=680) | All-Cause Mortality (n=414) | No Event (n=1790) |
|--------------------------------------|-----------------------|------------------------|-----------------------------|-------------------|
| Male (%)                             | 1680 (58)             | 442 (65)               | 273 (66)                    | 965 (54)          |
| <b>Ethnicity</b>                     |                       |                        |                             |                   |
| Asian                                | 201 (7)               | 56 (8)                 | 13 (3)                      | 132 (7)           |
| Black                                | 87 (3)                | 26 (4)                 | 8 (2)                       | 53 (3)            |
| Others                               | 79 (3)                | 10 (2)                 | 6 (1)                       | 63 (4)            |
| White                                | 2517 (87)             | 588 (87)               | 387 (94)                    | 1542 (86)         |
| Age, yr                              | 63 (15)               | 61 (15)                | 72 (10)                     | 61 (14)           |
| eGFR, ml/min per 1.73 m <sup>2</sup> | 35 (25–47)            | 22 (19–29)             | 32 (24–41)                  | 41 (32–53)        |
| Creatinine, mg/dl                    | 1.8 (1.4–2.4)         | 2.6 (2.1–3.1)          | 1.9 (1.5–2.4)               | 1.6 (1.3–2.0)     |
| UACR, mg/g                           | 197 (32–894.9)        | 858 (219–1983)         | 178 (27–652)                | 105 (22–540)      |
| BMI, kg/m <sup>2</sup>               | 28.7 (25.3–33.0)      | 28.8 (25.0–33.2)       | 29.6 (25.8–34.5)            | 28.5 (25.3–32.7)  |
| <b>Primary kidney diagnosis</b>      |                       |                        |                             |                   |
| CKD unknown cause                    | 961 (33)              | 208 (31)               | 161 (39)                    | 593 (33)          |
| Glomerular disease                   | 649 (23)              | 135 (20)               | 68 (16)                     | 447 (25)          |
| Diabetic kidney disease              | 326 (11)              | 113 (17)               | 81 (20)                     | 132 (7)           |
| Inherited                            | 292 (10)              | 93 (14)                | 11 (3)                      | 188 (11)          |
| Ischemic nephropathy                 | 256 (9)               | 58 (9)                 | 48 (12)                     | 150 (8)           |
| Tubulointerstitial disease           | 155 (5)               | 25 (4)                 | 12 (3)                      | 118 (7)           |
| Urological conditions                | 133 (5)               | 25 (4)                 | 13 (3)                      | 95 (5)            |
| Other systemic disease               | 71 (3)                | 14 (2)                 | 17 (4)                      | 40 (2)            |
| Congenital or dysplastic             | 41 (1.4)              | 9 (1.3)                | 3 (0.7)                     | 29 (2)            |
| <b>CKD stages</b>                    |                       |                        |                             |                   |
| 1                                    | 53 (2)                | 0 (0)                  | 3 (0.7)                     | 50 (3)            |
| 2                                    | 250 (9)               | 3 (0.4)                | 16 (4)                      | 231 (13)          |
| 3a                                   | 504 (18)              | 18 (3)                 | 53 (13)                     | 433 (24)          |
| 3b                                   | 979 (34)              | 133 (20)               | 152 (37)                    | 694 (39)          |
| 4                                    | 1098 (38)             | 526 (77)               | 190 (46)                    | 382 (21)          |
| Alcohol                              | 1012 (35)             | 208 (31)               | 64 (16)                     | 740 (41)          |
| <b>Smoking</b>                       |                       |                        |                             |                   |
| Never smoked                         | 1000 (35)             | 222 (33)               | 51 (12)                     | 727 (41)          |
| Ex-smoker                            | 842 (29)              | 190 (28)               | 91 (22)                     | 561 (31)          |
| Smoker                               | 134 (5)               | 37 (5)                 | 12 (3)                      | 85 (5)            |
| <b>Comorbidities</b>                 |                       |                        |                             |                   |
| Hypertension                         | 2404 (83)             | 629 (93)               | 335 (81)                    | 1440 (80)         |
| Diabetes                             | 879 (31)              | 228 (34)               | 208 (50)                    | 443 (25)          |
| Abdominal aortic aneurysm            | 9 (0.3)               | 4 (0.6)                | 3 (0.7)                     | 2 (0.1)           |
| Atrial fibrillation                  | 277 (10)              | 41 (6)                 | 90 (22)                     | 146 (8)           |
| Prior vascular disease <sup>a</sup>  | 599 (21)              | 148 (22)               | 177 (43)                    | 274 (15)          |
| <b>Medications</b>                   |                       |                        |                             |                   |
| ACEi                                 | 1178 (41)             | 288 (42)               | 140 (34)                    | 750 (42)          |
| ARB                                  | 890 (31)              | 233 (34)               | 97 (23)                     | 560 (31)          |
| Statins                              | 1704 (59)             | 445 (65)               | 267 (65)                    | 992 (55)          |

Data are presented as number (%) for categorical variables, mean for normally distributed continuous variables (age), and median (25th–75th percentile) for skewed continuous variables. Alcohol defined as self-reported regular intake exceeding 14 units per week. ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BMI, body mass index; UACR, urinary albumin-to-creatinine ratio.

<sup>a</sup>Prior vascular disease is a composite of coronary revascularization procedures, myocardial infarction, stroke, peripheral vascular disease, and amputation. Censored refers to participants with CKD who did not experience kidney failure or all-cause mortality from baseline to last follow-up. Kidney failure was defined as the first event of sustained eGFR <15 ml/min per 1.73 m<sup>2</sup> or commencement of KRT. Death was defined as death from any cause before developing kidney failure. Missing data were as follows: urinary albumin-to-creatinine ratio 248 (8.6%), body mass index 77 (2.7%), alcohol 919 (31.9%), smoking 908 (31.5%), hypertension 59 (2.0%), diabetes 59 (2.0%), abdominal aortic aneurysm 59 (2.0%), atrial fibrillation 59 (2.0%), prior vascular disease 295 (10.2%), angiotensin-converting enzyme inhibitor 40 (1.4%), angiotensin receptor blocker 40 (1.4%), and statins 40 (1.4%).

biomarkers according to C-index ranking is illustrated in Figure 3, A–C, for kidney failure, all-cause mortality, and the composite of the two. The optimal models (optimizing the balance between simplicity and performance) included the following biomarkers:

kidney failure: sTNFR1, sCD40, and UCOL1 (C-index, 0.86; 95% CI, 0.83 to 0.89); all-cause mortality: hs-cTnT, NT-proBNP, and suPAR (C-index, 0.80; 95% CI, 0.75 to 0.84); composite (kidney failure and all-cause mortality): sTNFR1, GDF-15, hs-cTnT, NT-proBNP, NGAL, and UCOL1A1 (C-index, 0.78; 95% CI, 0.76 to 0.81).

The C-index of risk prediction models, including established risk factors and novel biomarkers alone and in combination, is presented in Table 3.

The kidney failure risk score that included three key biomarkers alone achieved good discrimination (C-index 0.86) that was superior to UACR alone (0.71) and only slightly lower than eGFR (0.87) and eGFR and UACR combined (0.90), and in combination with sex+age+White ethnicity (0.90). Adding the biomarkers to the established risk factor model did not improve discrimination (*P* value 0.479; Table 4).

Table 2. Biomarker concentrations in participants with CKD and controls

| Biomarker           | All Control (n=86)   | Control without Diabetes (n=63) | Control with Diabetes (n=23)       | All CKD (n=2884)                      | CKD Stage G1 (n=53)    | CKD Stage G2 (n=250)   | CKD Stage G3a (n=504)  | CKD Stage G3b (n=979)  | CKD Stage G4 (n=1098)               |
|---------------------|----------------------|---------------------------------|------------------------------------|---------------------------------------|------------------------|------------------------|------------------------|------------------------|-------------------------------------|
| KIM-1, pg/ml        | 66.5 (37.6–154.7)    | 62.2 (30.3–154.7)               | 85.6 (59.4–141.1)                  | 250.8 (151.1–447.6) <sup>a,b</sup>    | 167.0 (72.6–267.5)     | 182.4 (100.4–303.4)    | 192.7 (117.5–309.6)    | 256.6 (154.7–449.2)    | 316.3 (194.8–534.0) <sup>b,c</sup>  |
| UKIM-1, ng/mg       | 103.8 (65.8–144.0)   | 86.6 (52.0–128.4)               | 227.5 (105.2–268.1) <sup>d,e</sup> | 288.4 (144.1–559.4) <sup>a,b</sup>    | 306.9 (162.6–501.2)    | 246.0 (116.4–521.0)    | 236.3 (114.0–471.9)    | 287.4 (139.0–570.5)    | 316.0 (167.3–634.8) <sup>b,c</sup>  |
| NGAL, ng/ml         | 95.1 (81.4–135.2)    | 92.3 (77.6–134.7)               | 95.3 (87.8–132.1)                  | 247.0 (172.9–348.5) <sup>a,b</sup>    | 112.9 (94.3–152.1)     | 159.4 (117.3–202.8)    | 184.9 (137.4–233.7)    | 236.6 (176.4–315.6)    | 331.4 (253.4–456.8) <sup>b,c</sup>  |
| Galectin-3, ng/ml   | 13.3 (9.6–20.9)      | 12.7 (9.6–18.9)                 | 18.1 (12.9–22.7)                   | 24.5 (15.4–34.9) <sup>a,b</sup>       | 16.3 (11.2–25.8)       | 19.1 (12.9–28.9)       | 20.2 (13.8–29.1)       | 23.7 (16.2–33.6)       | 28.5 (19.3–40.7) <sup>b,c</sup>     |
| suPAR, ng/ml        | 2.8 (2.3–3.3)        | 2.6 (2.1–3.1)                   | 3.0 (2.8–3.7) <sup>d</sup>         | 5.0 (3.9–6.5) <sup>a,b</sup>          | 3.2 (2.6–4.0)          | 3.5 (2.9–4.5)          | 4.1 (3.3–5.1)          | 4.9 (4.0–6.1)          | 6.0 (5.0–7.5) <sup>b,c</sup>        |
| UClusterin, μg/mg   | 8.1 (5.1–12.8)       | 7.5 (4.6–11.4)                  | 10.0 (8.0–16.5) <sup>d</sup>       | 22.4 (10.6–50.3) <sup>a,b</sup>       | 17.6 (9.0–42.1)        | 17.5 (8.9–42.3)        | 17.0 (7.5–35.8)        | 20.7 (9.9–47.4)        | 30.4 (13.6–65.2) <sup>b,c</sup>     |
| UOsteactivin, ng/mg | 261.9 (174.3–335.9)  | 242.7 (167.3–312.6)             | 322.4 (219.1–458.9)                | 405.2 (274.2–595.3) <sup>a,b</sup>    | 360.9 (249.9–563.0)    | 353.9 (237.1–485.8)    | 334.4 (224.0–511.2)    | 411.3 (281.9–583.2)    | 454.5 (314.6–651.8) <sup>b,c</sup>  |
| UCalbindin, ng/mg   | 579.2 (470.4–892.1)  | 565.8 (462.5–808.8)             | 792.6 (498.8–1069.6)               | 543.1 (351.4–855.5) <sup>a</sup>      | 860.0 (557.7–1332.4)   | 713.1 (486.6–1082.1)   | 663.0 (433.6–949.4)    | 535.6 (357.1–859.7)    | 455.4 (290.0–730.8) <sup>b,c</sup>  |
| UVEGF, ng/mg        | 76.9 (61.2–93.4)     | 78.0 (61.3–93.3)                | 75.2 (60.1–98.9)                   | 140.2 (74.5–267.3) <sup>a</sup>       | 111.6 (72.4–218.5)     | 133.3 (65.3–261.7)     | 118.3 (65.4–236.8)     | 136.2 (73.1–255.1)     | 154.6 (82.5–282.4) <sup>b,c</sup>   |
| UCOL1A1, ng/mg      | 134.5 (77.0–194.2)   | 125.6 (63.3–195.6)              | 155.2 (103.4–185.1)                | 410.9 (205.6–987.3) <sup>a,b</sup>    | 341.6 (165.0–555.5)    | 252.0 (144.9–496.1)    | 247.8 (153.4–490.8)    | 361.8 (192.9–769.8)    | 739.6 (338.8–1571.0) <sup>b,c</sup> |
| UTIMP1, ng/mg       | 41.3 (31.3–95.0)     | 33.8 (31.3–65.8)                | 57.3 (31.7–170.2)                  | 242.5 (131.8–469.6) <sup>a,b</sup>    | 147.1 (92.2–232.0)     | 168.9 (95.4–284.7)     | 157.3 (94.1–287.5)     | 222.3 (135.2–410.6)    | 356.1 (199.7–716.2) <sup>b,c</sup>  |
| UMMP-9, pg/mg       | 900.0 (900.0–1080.0) | 900.0 (900.0–1070.0)            | 900.0 (0.0–1765.0)                 | 2273.7 (1240.8–5308.8) <sup>a,b</sup> | 1760.8 (1273.5–4006.2) | 2698.3 (1221.6–5702.9) | 2273.7 (1187.7–4928.3) | 2118.7 (1187.7–4421.0) | 2340.5 (1328.8–7571.3)              |
| sCD40, pg/ml        | 94.1 (78.5–105.5)    | 87.3 (74.5–98.7)                | 102.0 (94.3–114.0) <sup>d</sup>    | 211.0 (156.3–285.7) <sup>a,b</sup>    | 107.8 (81.2–128.7)     | 129.6 (105.2–168.8)    | 155.3 (124.1–191.2)    | 203.9 (164.1–257.1)    | 283.9 (228.4–363.8) <sup>b,c</sup>  |
| sCD40L, pg/ml       | 58.5 (43.6–94.5)     | 58.5 (42.0–96.8)                | 58.5 (46.2–86.0)                   | 86.3 (57.5–135.4)                     | 89.4 (55.2–141.3)      | 88.8 (55.1–136.0)      | 81.9 (56.6–132.6)      | 86.4 (57.7–137.0)      | 87.4 (58.2–135.7)                   |
| sTNFR1, ng/ml       | 1.1 (0.9–1.3)        | 1.0 (0.8–1.3)                   | 1.2 (1.1–1.3)                      | 3.4 (2.4–4.7) <sup>a,b</sup>          | 1.4 (1.0–1.9)          | 1.9 (1.5–2.5)          | 2.4 (1.8–3.0)          | 3.2 (2.5–4.1)          | 4.7 (3.7–5.9) <sup>b,c</sup>        |
| CRP, mg/L           | 1.4 (0.7–2.8)        | 1.3 (0.7–2.7)                   | 1.8 (0.9–3.3)                      | 2.5 (1.1–5.6) <sup>a,b</sup>          | 1.4 (0.5–3.9)          | 2.1 (0.8–5.0)          | 2.0 (0.9–4.5)          | 2.6 (1.2–5.5)          | 2.9 (1.2–6.6) <sup>b,c</sup>        |
| MCP-1, pg/ml        | 175.4 (135.3–220.4)  | 169.7 (127.3–213.5)             | 196.1 (172.5–225.9)                | 237.3 (189.0–292.0) <sup>a,b</sup>    | 204.2 (145.6–244.5)    | 215.3 (169.0–271.9)    | 222.4 (181.5–274.3)    | 239.2 (190.4–297.4)    | 245.8 (200.6–303.8) <sup>b,c</sup>  |
| hs-cTnT, ng/L       | 3.7 (3.0–6.9)        | 3.6 (3.0–6.4)                   | 4.7 (3.0–7.7)                      | 16.7 (9.3–30.4) <sup>a,b</sup>        | 4.2 (3.0–6.8)          | 8.5 (4.7–13.7)         | 9.7 (5.8–17.3)         | 15.8 (10.0–27.7)       | 26.4 (16.2–43.4) <sup>b,c</sup>     |
| NT-proBNP, ng/L     | 51.9 (32.5–85.0)     | 49.6 (30.3–76.7)                | 56.9 (35.5–140.0)                  | 237.0 (98.1–663.8) <sup>a,b</sup>     | 48.3 (26.6–113.0)      | 89.2 (52.4–193.0)      | 136.0 (59.7–293.5)     | 216.5 (96.8–617.2)     | 459.0 (207.0–1122.0) <sup>b,c</sup> |

Table 2. Continued

| Biomarker     | All Control (n=86)  | Control without Diabetes (n=63) | Control with Diabetes (n=23)       | All CKD (n=2884)                      | CKD Stage G1 (n=53)  | CKD Stage G2 (n=250)  | CKD Stage G3a (n=504)  | CKD Stage G3b (n=979)  | CKD Stage G4 (n=1098)                 |
|---------------|---------------------|---------------------------------|------------------------------------|---------------------------------------|----------------------|-----------------------|------------------------|------------------------|---------------------------------------|
| GDF-15, pg/ml | 619.5 (504.0–976.8) | 583.0 (482.0–749.5)             | 1039.0 (852.5–1702.5) <sup>d</sup> | 2437.0 (1575.0–3748.0) <sup>a,b</sup> | 812.0 (548.0–1315.0) | 1354.0 (936.5–2072.5) | 1612.0 (1108.0–2373.0) | 2364.0 (1681.2–3519.0) | 3365.5 (2379.2–4647.8) <sup>b,c</sup> |
| FGF-23, pg/ml | 89.8 (68.5–115.6)   | 90.7 (69.4–114.2)               | 86.3 (68.3–114.9)                  | 199.1 (130.6–322.0) <sup>a,b</sup>    | 124.1 (97.7–175.7)   | 122.8 (90.0–191.3)    | 140.0 (104.3–207.6)    | 188.0 (130.6–283.9)    | 277.6 (190.9–425.8) <sup>b,c</sup>    |

Biomarker concentrations in participants with CKD and controls. Median (25th–75th percentiles). For urinary biomarkers (urinary kidney injury molecule-1, urinary clusterin, urinary osteoactivin, urinary calbindin, urinary vascular endothelial growth factor, urinary collagen type 1  $\alpha$ 1 chain, urinary tissue inhibitor of metalloproteinases 1, and urinary matrix metalloproteinase 9), concentrations are presented as values normalized for urinary creatinine (mg/dl). Superscript letter outside the brackets of the biomarkers' concentrations indicate a statistically significant difference when compared with a comparator group: To account for multiple comparisons, *P* values were adjusted using Bonferroni correction. CRP, C-reactive protein; FGF-23, fibroblast growth factor 23; GDF-15, growth differentiation factor-15; hs-cTnT, high-sensitivity cardiac troponin T; KIM-1, kidney injury molecule-1; MCP-1, monocyte chemoattractant protein-1; NGAL, neutrophil gelatinase-associated lipocalin; NT-proBNP, N-terminal pro-brain natriuretic peptide; sCD40, soluble cluster of differentiation 40; sCD40L, soluble cluster of differentiation 40 ligand; sTNFR1, soluble TNF receptor 1; suPAR, soluble urokinase plasminogen activator receptor; U, urine biomarkers; UCalbindin, urinary calbindin; UClusterin, urinary clusterin; UCOL1A1, urinary collagen type 1  $\alpha$ 1 chain; UKIM-1, urinary kidney injury molecule-1; UMMP-9, urinary matrix metalloproteinase 9; UOsteoactivin, urinary osteoactivin; UTIMP1, urinary tissue inhibitor of metalloproteinases 1; UVEGF, urinary vascular endothelial growth factor.

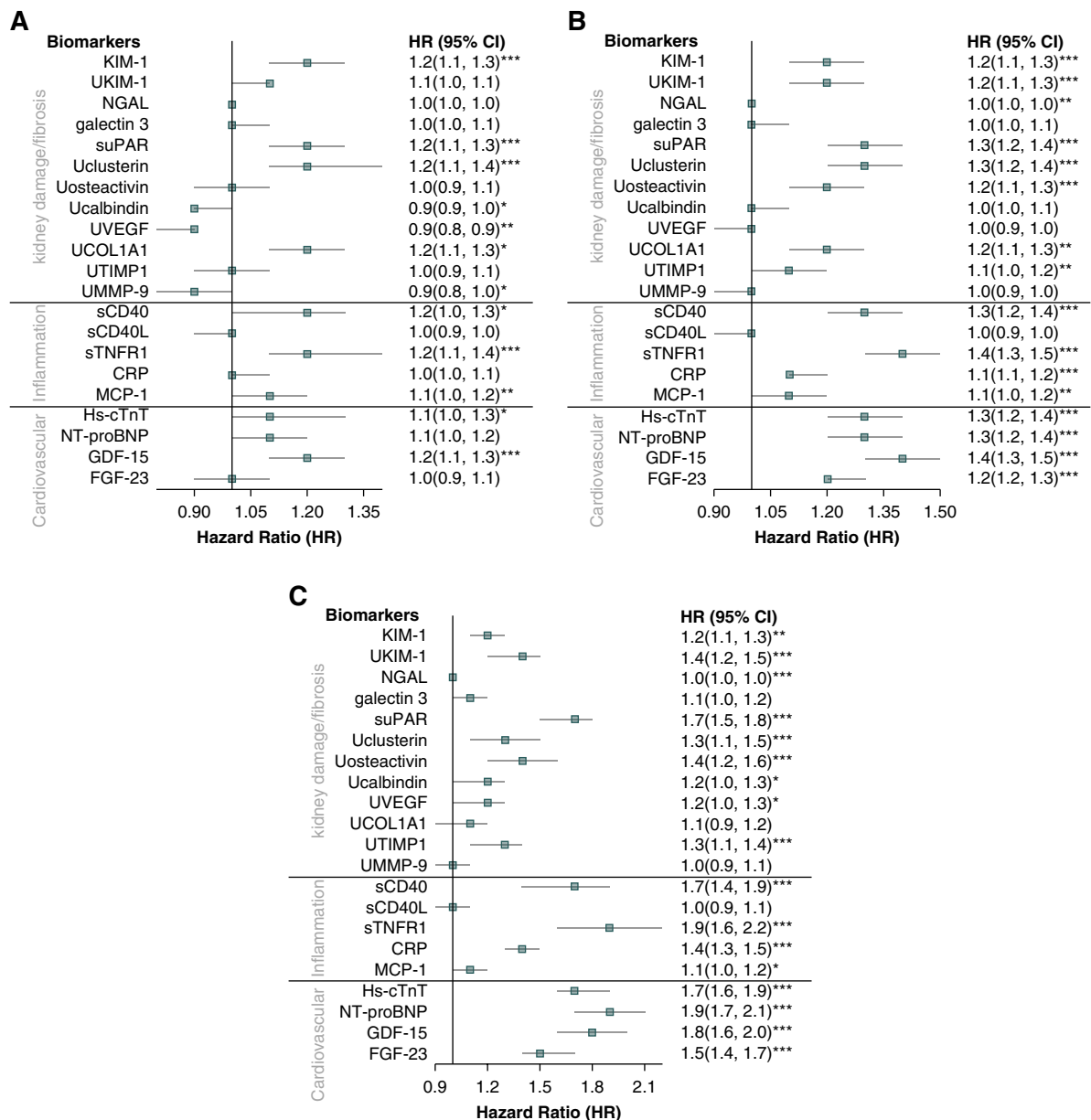
<sup>a</sup>All CKD versus all controls.

<sup>b</sup>Where: *P* value < 0.05 (letter only), < 0.0001.

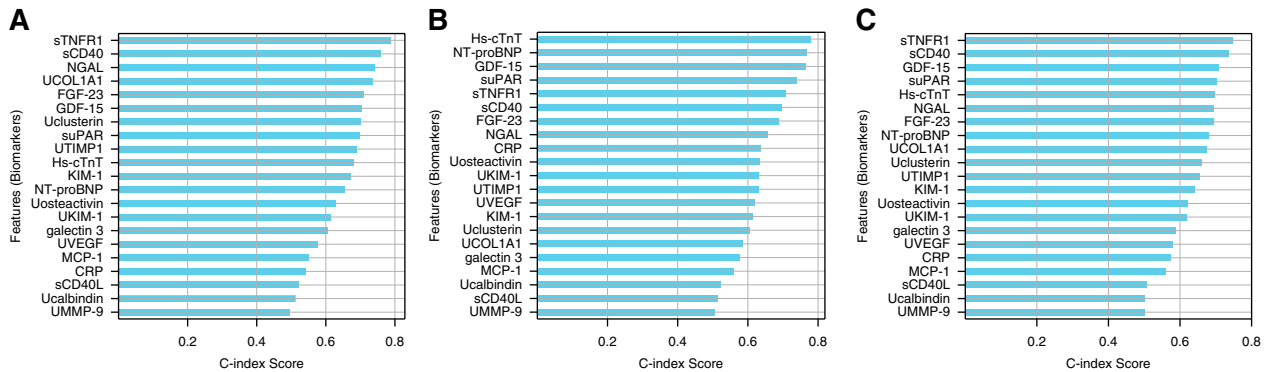
<sup>c</sup>Trend across CKD stages G1–4 compared with all CKD.

<sup>d</sup>Controls with diabetes versus controls without diabetes.

<sup>e</sup>Where: *P* value < 0.05 (letter only), < 0.001.



**Figure 1. Multiple biomarkers representing pathological mechanisms are independently associated with higher risks of kidney failure, all-cause mortality, and their composite outcome.** Adjusted hazard ratios of biomarkers for (A) kidney failure, (B) all-cause mortality, and (C) the composite outcome of kidney failure or all-cause mortality. Hazard ratios are expressed per one SD higher level of each biomarker. HRs are expressed per one SD higher level of each biomarker. The HR represents the change in the risk of outcome associated with a one-SD higher log-transformed biomarker level. The asterisks outside the brackets of the CI indicate statistical significance:  $P$  value  $< 0.05$  (\*),  $0.001$  (\*\*), and  $0.0001$  (\*\*\*). The 95% CI of the lower and upper HR is displayed on the right side of the plot. For the event of kidney failure, biomarkers were adjusted for age, sex, White ethnicity, eGFR, and albumin-to-creatinine ratio. For all-cause mortality and the composite of kidney failure and all-cause mortality, further adjustments were made for IMD rank, smoking, alcohol, diabetes, hypertension, and prior vascular disease (which is a composite of coronary revascularization procedures), MI, stroke, peripheral vascular disease, and amputation, as well as statins and ACEi/ARB. ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; CI, confidence interval; CRP, C-reactive protein; FGF-23, fibroblast growth factor 23; Galectin-3, Galectin-3; GDF-15, growth differentiation factor-15; HR, hazard ratio; hs-cTnT, high-sensitivity cardiac troponin T; IMD, Index of Multiple Deprivation; KIM-1, kidney injury molecule-1; MCP-1, monocyte chemoattractant protein-1; MI, myocardial infarction; NGAL, neutrophil gelatinase-associated lipocalin; NT-proBNP, N-terminal pro-brain natriuretic peptide; sCD40, soluble cluster of differentiation 40; sCD40L, soluble cluster of differentiation 40 ligand; sTNFR1, soluble TNF receptor 1; suPAR, soluble urokinase plasminogen activator receptor; Ucalbindin, urinary calbindin; Uclusterin, urinary clusterin; UCOL1A1, urinary collagen type 1  $\alpha 1$  chain; UKIM-1, urinary kidney injury molecule-1; UMMP-9, urinary matrix metalloproteinase 9; Uosteoactivin, urinary osteoactivin; UTIMP1, urinary tissue inhibitor of metalloproteinases 1; UVEGF, urinary vascular endothelial growth factor. Figure 1 can be viewed in color online at [www.jasn.org](http://www.jasn.org).



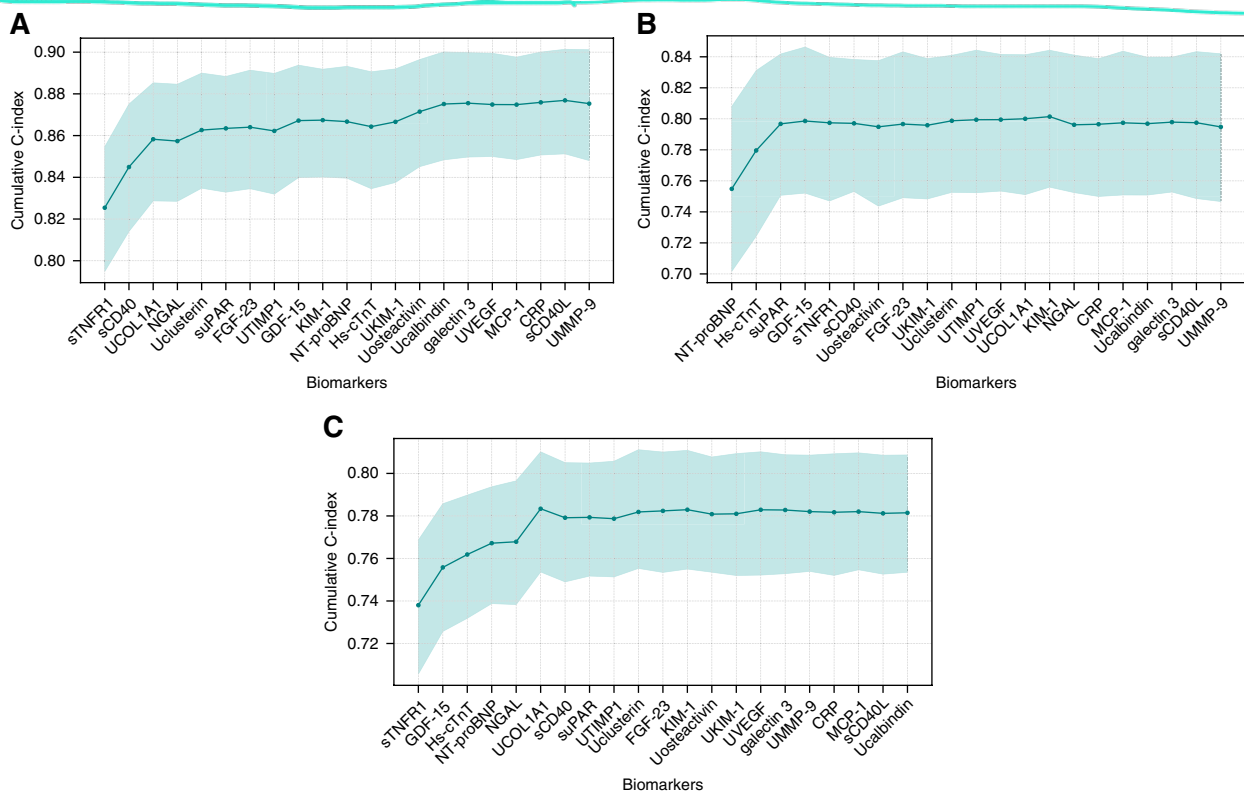
**Figure 2. Discrimination differs by biomarker and outcome.** Feature importance plots ranking biomarkers associated with (A) kidney failure, (B) all-cause mortality, and (C) the composite outcome of kidney failure or all-cause mortality, based on the C-index. Figure 2 can be viewed in color online at [www.jasn.org](http://www.jasn.org).

Regarding all-cause mortality, the biomarker-based model (C-index 0.80) performed equivalently to the established risk factor model (C-index 0.80). Combining the biomarker model with clinical factors resulted in a numerically higher C-index (0.83) with borderline statistical significance ( $P$  value = 0.054; Table 4).

For the composite outcome, the biomarker model discrimination (C-index 0.78) was numerically higher than for established risk factors (C-index 0.77), and the addition of biomarkers to the established risk factors led to a small but

statistically significant improvement in discrimination (C-index 0.80,  $P$  value <0.01).

Supplemental Tables 23 and 24 present the diagnostic performance measures for the established risk factor and combined biomarker models, treating the outcome as a binary (noncensored) end point. The number of persons needed to screen to detect one more in the top risk decile for each biomarker model versus established risk factors was 25, 11, and 19 for kidney failure, all-cause mortality, and the combined end point, respectively. The individual



**Figure 3. A small number of biomarkers is sufficient to achieve strong discrimination for kidney failure, all-cause mortality, and their composite outcome.** Cumulative C-index plots showing the iterative addition of biomarkers, ranked by C-index, for (A) kidney failure, (B) all-cause mortality, and (C) the composite outcome of kidney failure or all-cause mortality. Shading depicts the lower and upper CIs. Figure 3 can be viewed in color online at [www.jasn.org](http://www.jasn.org).

**Table 3. C-index of risk models incorporating established risk factors and novel biomarkers, individually and combined for the end points of kidney failure, all-cause mortality, and composite kidney failure or all-cause mortality**

| Covariates                                      | Kidney Failure<br>C-Index (95% CI) | Covariates                                                             | All-Cause Mortality<br>C-Index (95% CI) | Composite<br>C-Index (95% CI) |
|-------------------------------------------------|------------------------------------|------------------------------------------------------------------------|-----------------------------------------|-------------------------------|
| Sex                                             | 0.54 (0.49 to 0.58)                | Sex                                                                    | 0.51 (0.46 to 0.56)                     | 0.55 (0.51 to 0.58)           |
| Age                                             | 0.55 (0.49 to 0.60)                | Age                                                                    | 0.70 (0.65 to 0.75)                     | 0.59 (0.55 to 0.63)           |
| Ethnicity                                       | 0.50 (0.47 to 0.53)                | Ethnicity                                                              | 0.54 (0.51 to 0.56)                     | 0.50 (0.48 to 0.52)           |
| UACR                                            | 0.72 (0.67 to 0.76)                | Mortality risk factors                                                 | 0.77 (0.72 to 0.81)                     | 0.65 (0.61 to 0.68)           |
| eGFR                                            | 0.87 (0.84 to 0.89)                | UACR                                                                   | 0.56 (0.50 to 0.61)                     | 0.65 (0.62 to 0.69)           |
| Combined covariates                             |                                    | eGFR                                                                   | 0.63 (0.57 to 0.69)                     | 0.73 (0.70 to 0.76)           |
| a. Sex+age+ethnicity                            | 0.56 (0.50 to 0.61)                | Combined covariates                                                    |                                         |                               |
| b. UACR+eGFR                                    | 0.90 (0.87 to 0.92)                | a. Sex+age+ethnicity+<br>mortality risk factors <sup>a</sup>           | 0.79 (0.74 to 0.83)                     | 0.66 (0.62 to 0.69)           |
| a+b                                             | 0.90 (0.88 to 0.92)                | b. UACR+eGFR                                                           | 0.64 (0.58 to 0.70)                     | 0.76 (0.73 to 0.79)           |
| c. Biomarkers of<br>kidney failure <sup>b</sup> | 0.86 (0.83 to 0.89)                | a+b                                                                    | 0.80 (0.76 to 0.84)                     | 0.77 (0.74 to 0.80)           |
| (a+b)+c                                         | 0.90 (0.88 to 0.92)                | c. Biomarkers of all-cause<br>mortality <sup>c</sup>                   | 0.80 (0.75 to 0.84)                     |                               |
|                                                 |                                    | (a+b)+c                                                                | 0.83 (0.79 to 0.87)                     |                               |
|                                                 |                                    | d. Biomarkers of kidney<br>failure or all-cause mortality <sup>d</sup> |                                         | 0.78 (0.76 to 0.81)           |
|                                                 |                                    | (a+b)+d                                                                |                                         | 0.80 (0.77 to 0.82)           |

CI, confidence interval; UACR, urinary albumin-to-creatinine ratio.

<sup>a</sup>Mortality risk factors: (Index of Multiple Deprivation score, smoking, alcohol, diabetes, hypertension, prior vascular disease [composite of coronary revascularization procedures, myocardial infarction, stroke, peripheral vascular disease, and amputation], statins, and angiotensin-converting enzyme inhibitor/angiotensin receptor blocker).

<sup>b</sup>Soluble TNF receptor 1, soluble CD40, and urinary collagen type 1  $\alpha$ 1 chain.

<sup>c</sup>High-sensitivity cardiac troponin T, N-terminal pro-brain natriuretic peptide, and soluble urokinase plasminogen activator receptor.

<sup>d</sup>Soluble TNF receptor 1, growth differentiation factor-15, high-sensitivity cardiac troponin T, N-terminal pro-brain natriuretic peptide, neutrophil gelatinase-associated lipocalin, and urinary collagen type 1  $\alpha$ 1 chain.

C-indices of each biomarker for the three outcomes, stratified by eGFR group, are listed in [Supplemental Tables 25–27](#). The performance of each model for the three outcomes by eGFR group is presented in [Supplemental Table 28](#). The varying discrimination of individual biomarkers and models may reflect differences in the relative prognostic importance of specific biomarkers at different stages of CKD or differences in event rates across eGFR strata, such that lower eGFR groups experienced more events (see [Supplemental Table 29](#)).

## Discussion

We identified significant associations between multiple biomarkers and three overlapping pathophysiological

processes and end points of kidney failure and all-cause mortality in a large cohort of persons with CKD. By iteratively combining a small number of the highest performing biomarkers, we developed risk prediction models for kidney failure and all-cause mortality that achieved excellent discrimination. Importantly, these models were constructed exclusively from biomarkers that reflect mechanisms of CKD progression and cardiovascular disease, which may in the future help guide and monitor more personalized treatments.

Despite extensive research, newer biomarkers in CKD have not consistently improved risk prediction beyond what is provided by eGFR and albuminuria.<sup>17,18</sup> In fact, the KFRE model consistently demonstrates excellent predictive accuracy across diverse populations<sup>19</sup> and supports

**Table 4. Comparison of C-index for established risk factor models with and without biomarkers**

| Events                                | Established Risk Factors<br>C-Index (95% CI) | Established Risk Factors+<br>Biomarkers C-Index (95% CI) | P Values |
|---------------------------------------|----------------------------------------------|----------------------------------------------------------|----------|
| Kidney failure                        | 0.90 (0.88 to 0.92)                          | 0.90 (0.88 to 0.92)                                      | 0.48     |
| All-cause mortality                   | 0.80 (0.76 to 0.84)                          | 0.83 (0.79 to 0.87)                                      | 0.054    |
| Kidney failure or all-cause mortality | 0.77 (0.74 to 0.80)                          | 0.80 (0.77 to 0.82)                                      | <0.001   |

Kidney failure events. Established risk factors: sex+age+ethnicity+urinary albumin-to-creatinine ratio+eGFR. Biomarkers: soluble TNF receptor 1+soluble CD40+collagen type 1  $\alpha$ 1. All-cause mortality events. Established risk factors: sex+age+ethnicity+urinary albumin-to-creatinine ratio+eGFR+Index of Multiple Deprivation rank, smoking, alcohol, diabetes, hypertension, prior vascular disease (a composite of coronary revascularization procedures, myocardial infarction, stroke, peripheral vascular disease, and amputation), statins, and angiotensin-converting enzyme inhibitor/angiotensin receptor blocker. Biomarkers: high-sensitivity cardiac troponin T+N-terminal pro-brain natriuretic peptide+soluble urokinase plasminogen activator receptor, kidney failure or all-cause mortality events. Established risk factors are the same as all-cause mortality. Biomarkers: soluble TNF receptor 1+growth differentiation factor-15+high-sensitivity cardiac troponin T+N-terminal pro-brain natriuretic peptide+neutrophil gelatinase-associated lipocalin+collagen type 1  $\alpha$ 1. P value compares C-index of two models. CI, confidence interval.

clinical risk stratification.<sup>3</sup> While incorporating these novel biomarkers into a model alongside established risk factors did not lead to improvement in risk prediction, this is perhaps unsurprising given the high discriminatory performance of established risk factors demonstrated in this and other studies.<sup>10</sup> Pursuing incremental gains in risk prediction alone may therefore be of limited clinical utility. However, the key limitation of the established risk factor model lies in its inability to capture ongoing disease activity and underlying mechanisms. As observed in our analysis, most of the discriminatory power in the established risk factor model was driven by eGFR (C-index 0.87), which primarily reflects the extent of previous kidney damage, as well as proximity to the end point (eGFR <15 ml/min).

By contrast, we have developed a risk prediction model with similar discriminatory performance to the KFRE using novel biomarkers alone. Although improvements in discrimination may not be necessary, this approach provides similar performance, but also reflects current disease activity and pathophysiological processes that may be responsive to kidney-protective treatments. Preclinical and other studies have established links between these biomarkers and underlying mechanisms; our findings support a potential role for them in the progression of human CKD, warranting further investigation. If confirmed, such models hold promise for informing a personalized approach to treatment, monitoring disease activity and response to treatment, and the development of novel therapies.<sup>11</sup>

After adjusting for established risk factors, 12 biomarkers were significantly associated with a higher risk of kidney failure. Of these, three were included in the parsimonious biomarker model: one kidney damage biomarker, UCOL1A1, and two inflammatory biomarkers, sTNFR1 and sCD40.

The accumulation of extracellular matrix (ECM), driven by excessive type 1 collagen production, is a critical driver of CKD development contributing to glomerulosclerosis and tubulointerstitial fibrosis.<sup>20</sup> Key biomarkers of ECM deposition include matrix metalloproteinases and collagen markers.<sup>21</sup> During CKD progression, COL1A1 expression increases, resulting in excessive production and accumulation of collagen type 1, which contributes to ECM accumulation and fibrosis.<sup>17</sup> In our study, higher UCOL1A1 levels were independently associated with a higher risk of kidney failure and were included in our biomarker-based prediction model. In addition, lower levels of MMP-9, a key enzyme for ECM remodeling,<sup>17</sup> were associated with higher kidney failure risk. These biomarkers may help identify individuals who could benefit from therapies targeting collagen production or turnover and may also be useful for monitoring treatment response.

Chronic inflammation also plays a pivotal role in the progression of both kidney damage and atherosclerosis,<sup>13</sup> particularly through the TNF pathway. Two inflammatory biomarkers, sCD40 and sTNFR1, were independently associated with a higher risk of kidney failure. Whereas sTNFR1 has been extensively studied in CKD and shown to predict both CKD progression and mortality,<sup>4,22</sup> the role of sCD40 is less well understood. sCD40 is membrane

glycoprotein belonging to the TNF receptor superfamily and is expressed in various cells, including proximal tubular cells.<sup>23</sup> However, its evaluation as a biomarker in CKD has been limited to small cohorts, with mixed results.<sup>24,25</sup> The association of these inflammatory biomarkers with kidney failure risk not only provides mechanistic insights into CKD progression but also highlights potential therapeutic targets. Targeting inflammation in CKD is an appealing strategy for several reasons. First, inflammation is a well-established driver of CKD progression.<sup>13</sup> Second, many anti-inflammatory medications with proven safety profiles have been tested in CKD populations for other indications and could be repurposed CKD treatment.<sup>26</sup> Furthermore, several novel CKD treatments designed to inhibit inflammation have shown promising results in animal models.<sup>27</sup>

Three biomarkers were included in the best performing risk prediction model for all-cause mortality. These included established cardiovascular disease biomarkers (hs-cTnT and NT-proBNP) and the kidney damage and inflammatory biomarker suPAR. This is consistent with observations that cardiovascular events are the leading cause of mortality in CKD and that inflammation plays a significant role in the pathogenesis of cardiovascular disease.<sup>28</sup> hs-cTnT and NT-proBNP reflect myocardial cell injury<sup>29</sup> and ventricular wall distention,<sup>30</sup> respectively, and were the strongest predictors of all-cause mortality. Both have previously been linked to greater cardiovascular mortality risk in CKD.<sup>31</sup> suPAR has also been previously shown to be associated with higher mortality risk in CKD.<sup>19</sup> Given the strong association between inflammation and cardiovascular events, there is growing interest in using anti-inflammatory drugs, such as clazakizumab, an IL-6 inhibitor, to reduce cardiovascular events in CKD.<sup>32</sup> In this context, inflammatory biomarkers not only highlight potential therapeutic targets but may also identify those most likely to respond to treatment and provide a means for monitoring treatment response.

NURTURE-CKD is a large prospective cohort of individuals with nondialysis CKD that includes participants with various causes of CKD, making our findings broadly applicable to the wider CKD population cared for in nephrology clinics. All baseline biomarkers, including eGFR and UACR, were measured in a central laboratory to ensure consistency. Furthermore, the collection of routine laboratory data and clinical end points through the UK Renal Registry provided robust and comprehensive follow-up data.

Several limitations should be noted. First, data on cardiovascular events and cause of death are not yet available. Second, these findings may not be generalizable to populations that differ significantly from this cohort, such as patients with earlier stages of CKD, those managed in primary care settings, or populations with a different ethnic composition. Notably, 87% of the cohort are White and only 3% are Black, which is significant given the substantially higher risk of kidney failure among Black patients. Third, because of the modest number of events in participants with eGFR >45 ml/min per 1.73 m<sup>2</sup> over the 48-month follow-up, we were unable to develop robust biomarker models stratified by baseline eGFR. Fourth, we assessed only a subset of a

growing number of novel biomarkers. Finally, biomarker values were not disaggregated by sex, which is important given the established sex-based differences in biomarker distribution and CKD outcomes. A planned additional publication will specifically address sex-stratified analyses.

By examining a broad range of biomarkers associated with CKD progression mechanisms and cardiovascular disease, we have developed parsimonious risk prediction models for kidney failure, all-cause mortality, and a composite of the two, with excellent discriminatory performance. Discrimination was comparable with established risk factor models, while crucially providing valuable insights into mechanisms of progression. If validated in other populations, these biomarkers and models could be used to guide and monitor more personalized therapy. Further research is needed to evaluate how these biomarkers respond to current and novel treatments and their clinical application in personalized medicine.

#### Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at <http://links.lww.com/JSN/F289>.

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#### Declarative Statements

This study includes clinical experimentation and received Institutional Review Board or Ethics Committee approval. All patients provided written informed consent. This study includes clinical experimentation and complies with the Declaration of Helsinki.

#### Data Availability Statements

Partial restrictions to the data and/or materials apply. The data that support the findings of this study are not publicly available due to the requirements of the UK Data Protection Act and conditions of the Research Ethics Committee approval. Access to anonymized participant-level data will be made available to external investigators upon successful application to the independent data access committee, which can be made *via* the NURTURE website: <https://nurturebiobank.org/information-for-researchers/applying-for-data-and-samples/>. The data used were pseudonymized records from adult CKD participants from the NURTURE biorepository database. Statistical code and clinical code sets can be provided upon request.

#### Supplemental Material

This article contains the following supplemental material online at <http://links.lww.com/JSN/F290>

**Supplemental Table 1.** Information on included biomarkers and number of patients.

**Supplemental Table 2.** Biomarker analytical precision across serum/plasma and urine samples.

**Supplemental Table 3.** Biomarker levels in the control group, censored (CKD cohort with no events) versus CKD cohort with outcome events—kidney failure and all-cause mortality.

**Supplemental Table 4.** HRs from Cox proportional hazards models for the end point kidney failure: all participants with CKD.

**Supplemental Table 5.** HRs from Cox proportional hazards models for the kidney failure: participants with eGFR >45 ml/min per 1.73 m<sup>2</sup> (*n*=807).

**Supplemental Table 6.** HRs from Cox proportional hazards models for the kidney failure: participants with eGFR 30–45 ml/min per 1.73 m<sup>2</sup> (*n*=980).

**Supplemental Table 7.** HRs from Cox proportional hazards models for the end point kidney failure: participants with eGFR <30 ml/min per 1.73 m<sup>2</sup> (*n*=1097).

**Supplemental Table 8.** HRs from Cox proportional hazards models for the end point all-cause mortality: all participants with CKD.

**Supplemental Table 9.** HRs from Cox proportional hazards models for the end point all-cause mortality: participants with eGFR >45 ml/min per 1.73 m<sup>2</sup> (*n*=807).

**Supplemental Table 10.** HRs from Cox proportional hazards models for the end point all-cause mortality: participants with eGFR 30–45 ml/min per 1.73 m<sup>2</sup> (*n*=980).

**Supplemental Table 11.** HRs from composite Cox proportional hazards models for the endpoint all-cause mortality: participants with eGFR <30 ml/min per 1.73 m<sup>2</sup> (*n*=1097).

**Supplemental Table 12.** HRs from composite Cox proportional hazards models for the kidney failure or all-cause mortality end point: all participants with CKD.

**Supplemental Table 13.** HRs from composite Cox proportional hazards models for the kidney failure or end point all-cause mortality: participants with eGFR >45 ml/min per 1.73 m<sup>2</sup> (*n*=807).

**Supplemental Table 14.** HRs from composite Cox proportional hazards models for the kidney failure or end point all-cause mortality: participants with eGFR 30–45 ml/min per 1.73 m<sup>2</sup> (*n*=980).

**Supplemental Table 15.** HRs from composite Cox proportional hazards models for the kidney failure or end point all-cause mortality: participants with eGFR <30 ml/min per 1.73 m<sup>2</sup> (*n*=1097).

**Supplemental Table 16.** Comparison of linear and natural cubic spline Cox regression model fit for biomarker association with kidney failure events.

**Supplemental Table 17.** Comparison of linear and natural cubic spline Cox regression model fit for biomarker association with all-cause mortality events.

**Supplemental Table 18.** Model validation of biomarkers associated with kidney failure events.

**Supplemental Table 19.** Model validation of biomarkers associated with all-cause mortality events.

**Supplemental Table 20.** Model validation of biomarkers associated with kidney failure or all-cause mortality events.

**Supplemental Table 21.** Summary of training and validation datasets for kidney failure analysis.

**Supplemental Table 22.** Summary of training and validation datasets for all-cause mortality analysis.

**Supplemental Table 23.** Model performance metrics and comparison of discrimination.

**Supplemental Table 24.** Number needed to screen for the combined biomarker model compared with the established risk factor model.

**Supplemental Table 25.** Whole CKD cohort C-index of unadjusted biomarker models associated with kidney failure events.

**Supplemental Table 26.** Whole CKD cohort C-index of unadjusted biomarker models associated with all-cause mortality events.

**Supplemental Table 27.** Whole CKD cohort C-index of unadjusted biomarker models associated with kidney failure or all-cause mortality events.

**Supplemental Table 28.** Comparison of C-index from whole CKD cohort for risk models of key predictive biomarkers.

**Supplemental Table 29.** Summary of population, sample, and event rates in eGFR strata.

**Supplemental Figure 1.** Heatmap showing correlations among biomarkers, including eGFR and UACR by outcome.

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